

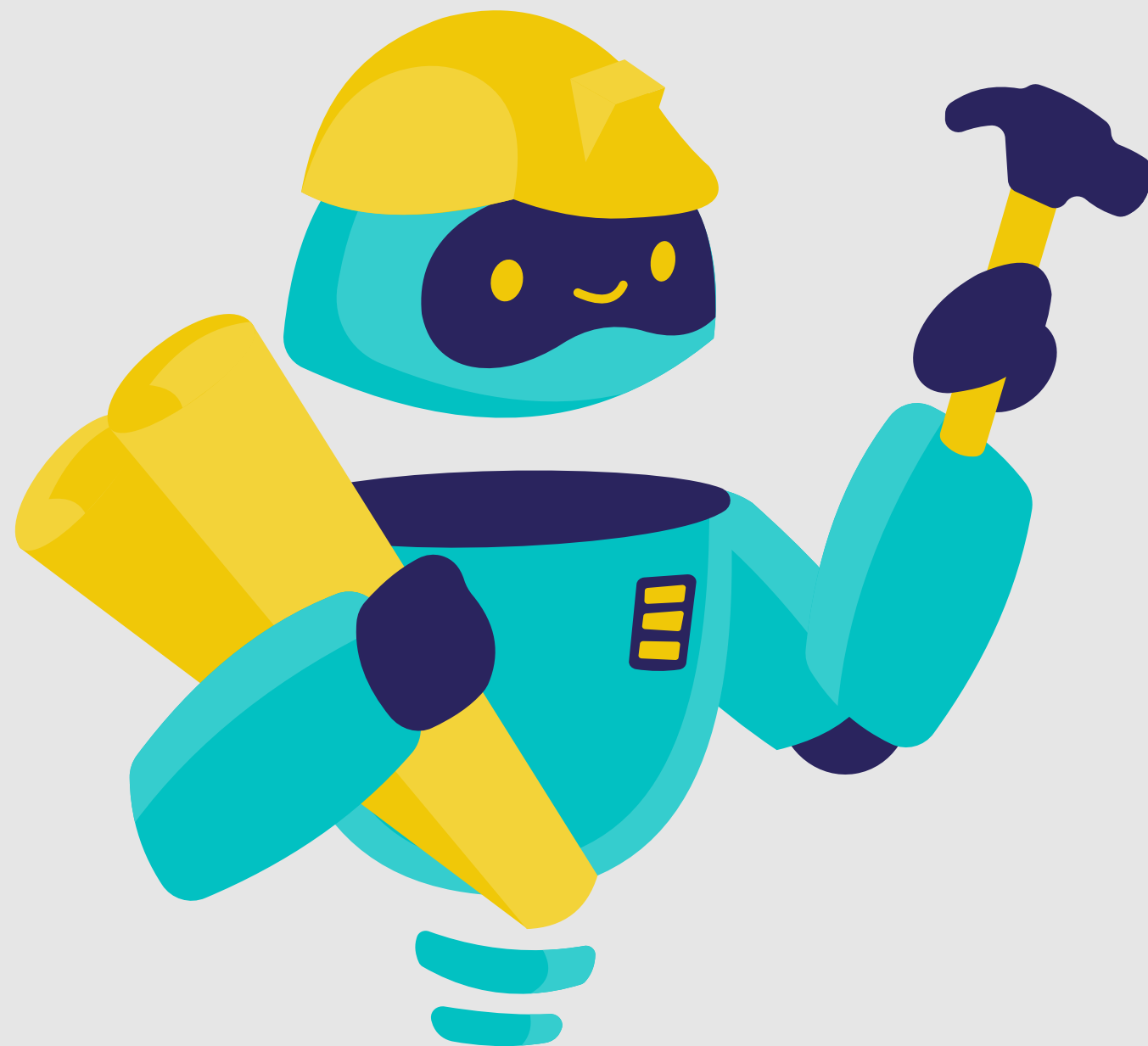


RENAD NASER TKROUNE

BUSINESS CASE: AUTOMATED CUSTOMER REVIEWS



INTRODUCTION



IN MANY COMPANIES, EVEN THOUGH CUSTOMER EXPERIENCE IS REGULARLY MEASURED, DATA TRANSFORMATION BECOMES CHALLENGING FOR MANY REASONS.

FOR INSTANCE, IT IS EXPENSIVE, RESOURCE-INTENSIVE, AND REQUIRES EXPERTISE. BUT THANKS TO MODERN TECHNOLOGIES LIKE MARTECH TOOLS AND THE USE OF ARTIFICIAL INTELLIGENCE (AI) IN CUSTOMER FEEDBACK ANALYSIS.

TYPES OF ARTIFICIAL INTELLIGENCE

- **SENTIMENT CLASSIFICATION (BERT-BASED)**

FINE-TUNED ON 15K REVIEWS TO CLASSIFY SENTIMENTS (E.G., POSITIVE, NEGATIVE)

- **REVIEW SUMMARIZATION (GPT-4)**

USES A TRANSFORMER TO GENERATE ARTICLE-STYLE SUMMARIES FOR EACH PRODUCT CATEGORY, HIGHLIGHTING TOP PRODUCTS, KEY FEATURES, AND MAJOR COMPLAINTS.

- **CLUSTERING MODEL (K-MEANS)**

GROUPS SIMILAR REVIEWS INTO CLUSTERS TO UNCOVER HIDDEN PATTERNS AND TOPICS—GREAT FOR EXPLORING THEMES WITHOUT LABELED DATA.



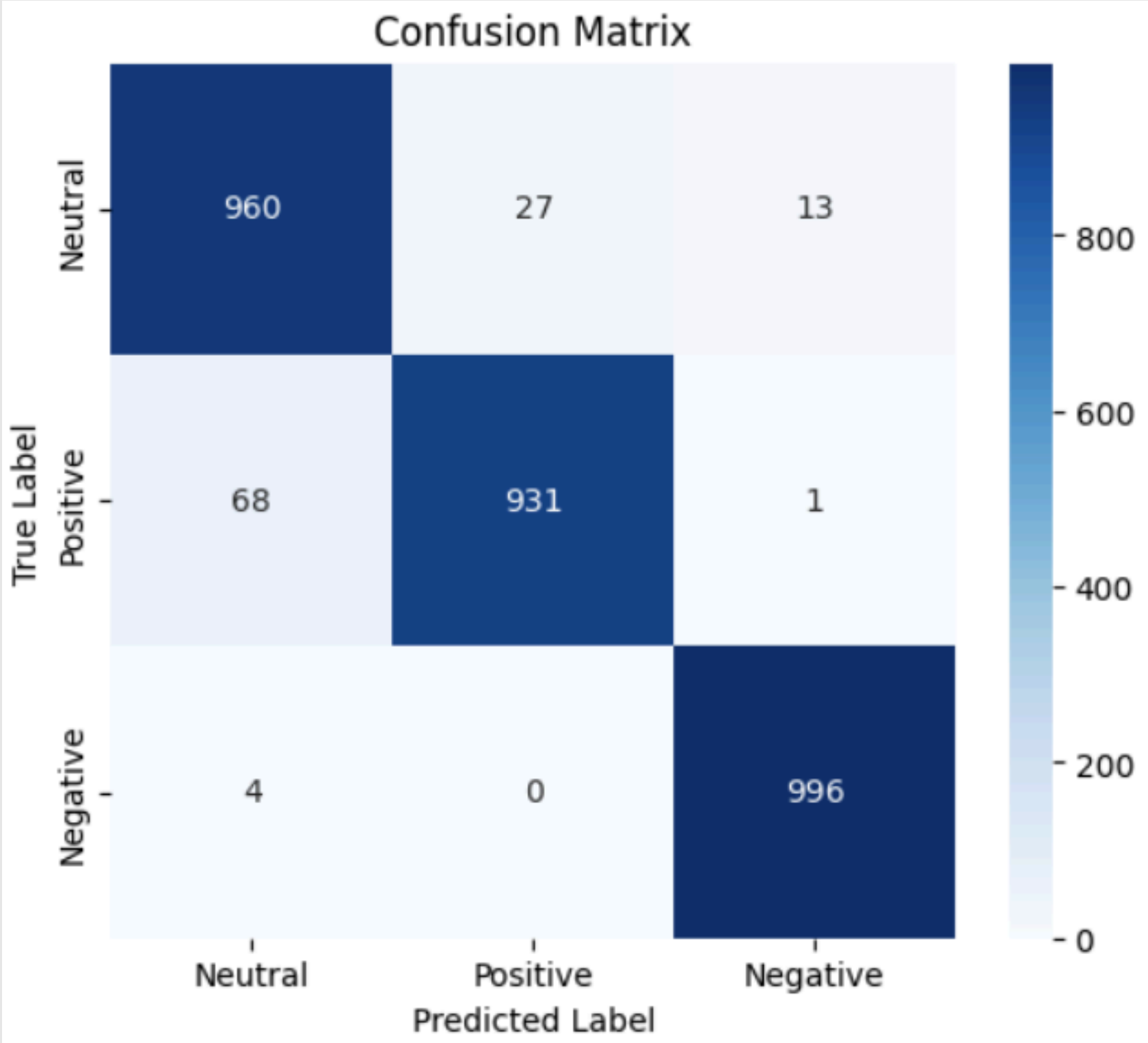
- EXPLORED DATASET STRUCTURE AND CONTENT TO UNDERSTAND FORMATS, DISTRIBUTIONS, AND POTENTIAL ISSUES
- REMOVED UNNECESSARY COLUMNS THAT DIDN'T CONTRIBUTE TO MODEL LEARNING
- COMBINED RELEVANT COLUMNS (TITLE + REVIEW TEXT) TO PROVIDE RICHER CONTEXT AND IMPROVE MODEL PERFORMANCE
- HANDLED NULL VALUES BY REMOVING ROWS OR REPLACING WITH MEANINGFUL DEFAULTS
- STANDARDIZED NUMERICAL VALUES TO ENSURE CONSISTENCY ACROSS ALL SAMPLES (RATINGS SCALED FROM 1-5)
- CLEANED TEXT DATA BY:
 - LOWERCASING ALL TEXT
 - REMOVING SPECIAL CHARACTERS, EXTRA WHITESPACE, AND STOP WORDS
 - HANDLING CONTRACTIONS (E.G., "DON'T" → "DO NOT")
- TOKENIZED AND ENCODED TEXT USING APPROPRIATE TOKENIZERS
- CHECKED CLASS BALANCE AND PERFORMED RESAMPLING IF NECESSAR



DATA PROCESSING & CLEANING

SENTIMENT CLASSIFICATION:

MODEL 1: DISTILBERT-BASE-UNCASED



- **MODEL:** DISTILBERT-BASE-UNCASED
- **DATASET:** 50,000 LABELED PRODUCT REVIEWS
- **CLASSES:** POSITIVE, NEGATIVE, NEUTRAL (MULTI-CLASS CLASSIFICATION)
- **ACCURACY:** 96% ON VALIDATION SET
- **PREPROCESSING:** TOKENIZATION WITH BERT TOKENIZER, LOWERCASING, SPECIAL TOKEN HANDLING
- **FINE-TUNING:** USED PRE-TRAINED BERT BASE MODEL WITH CLASSIFICATION HEAD
- **OPTIMIZER:** LEARNING RATE ADJUSTMENT
- **USE CASE:** AUTOMATES SENTIMENT TAGGING FOR REVIEWS, USEFUL IN DASHBOARDS, ALERTS, AND ANALYTICS

SENTIMENT CLASSIFICATION:

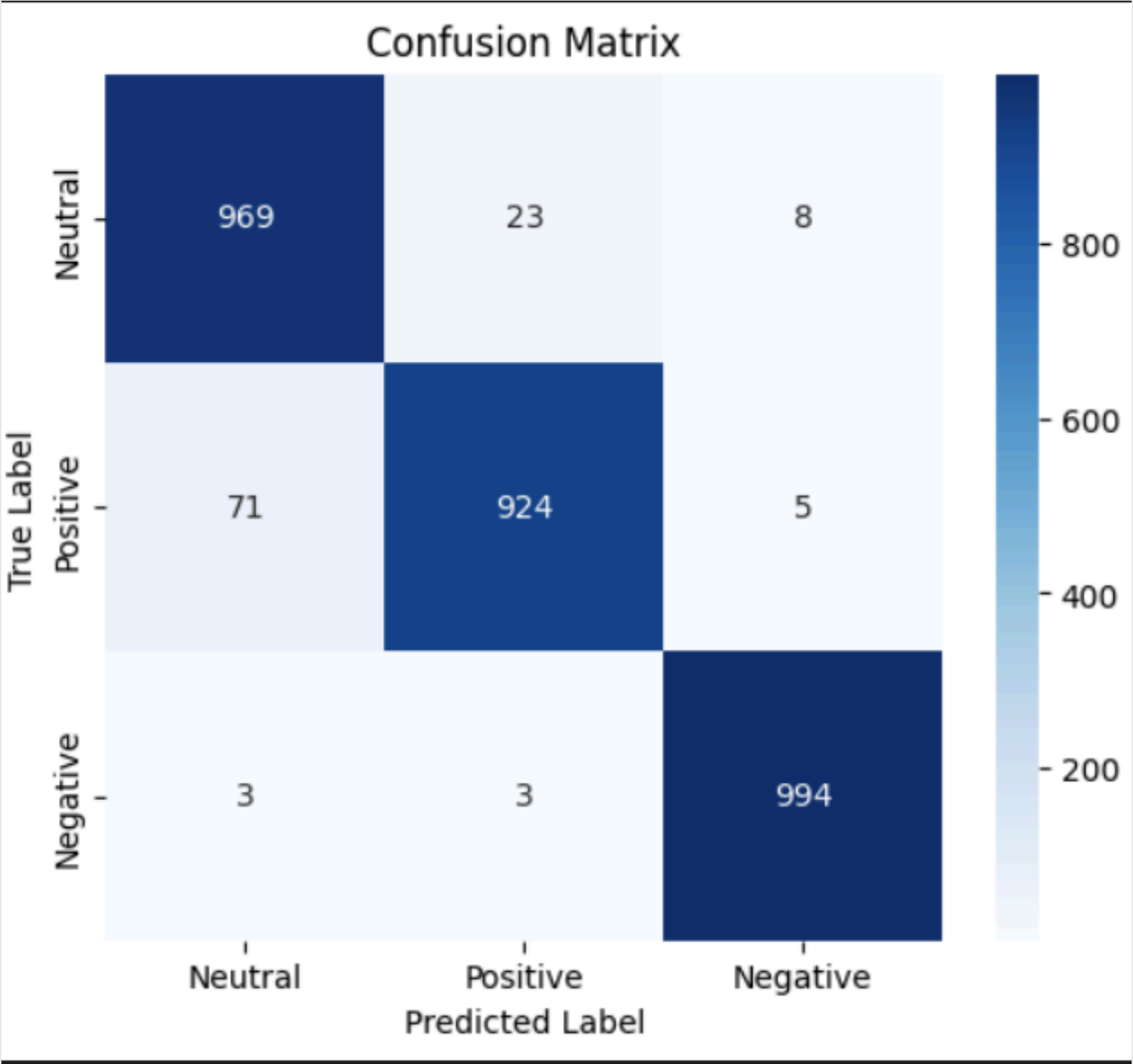
MODEL 1: DISTILBERT-BASE-UNCASED

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.589500	0.296927	0.902000	0.901968	0.901958	0.902000
2	0.201000	0.202221	0.940333	0.940361	0.941638	0.940333
3	0.112300	0.169736	0.961333	0.961276	0.961283	0.961333
4	0.058900	0.172140	0.962333	0.962299	0.962729	0.962333

	precision	recall	f1-score	support
Neutral	0.93	0.96	0.94	1000
Positive	0.97	0.93	0.95	1000
Negative	0.99	1.00	0.99	1000
accuracy			0.96	3000
macro avg	0.96	0.96	0.96	3000
weighted avg	0.96	0.96	0.96	3000

SENTIMENT CLASSIFICATION:

MODEL 2: BERT-BASE-UNCASED



- **MODEL:** BERT-BASE-UNCASED
- **DATASET:** 50,000 LABELED PRODUCT REVIEWS
- **CLASSES:** POSITIVE, NEGATIVE, NEUTRAL (MULTI-CLASS CLASSIFICATION)
- **ACCURACY:** 96% ON VALIDATION SET
- **PREPROCESSING:** TOKENIZATION WITH BERT TOKENIZER, LOWERCASING, SPECIAL TOKEN HANDLING
- **FINE-TUNING:** USED PRE-TRAINED BERT BASE MODEL WITH CLASSIFICATION HEAD
- **OPTIMIZER:** LEARNING RATE AND EPOCH ADJUSTMENT
- **USE CASE:** AUTOMATES SENTIMENT TAGGING FOR REVIEWS, USEFUL IN DASHBOARDS, ALERTS, AND ANALYTICS

SENTIMENT CLASSIFICATION:

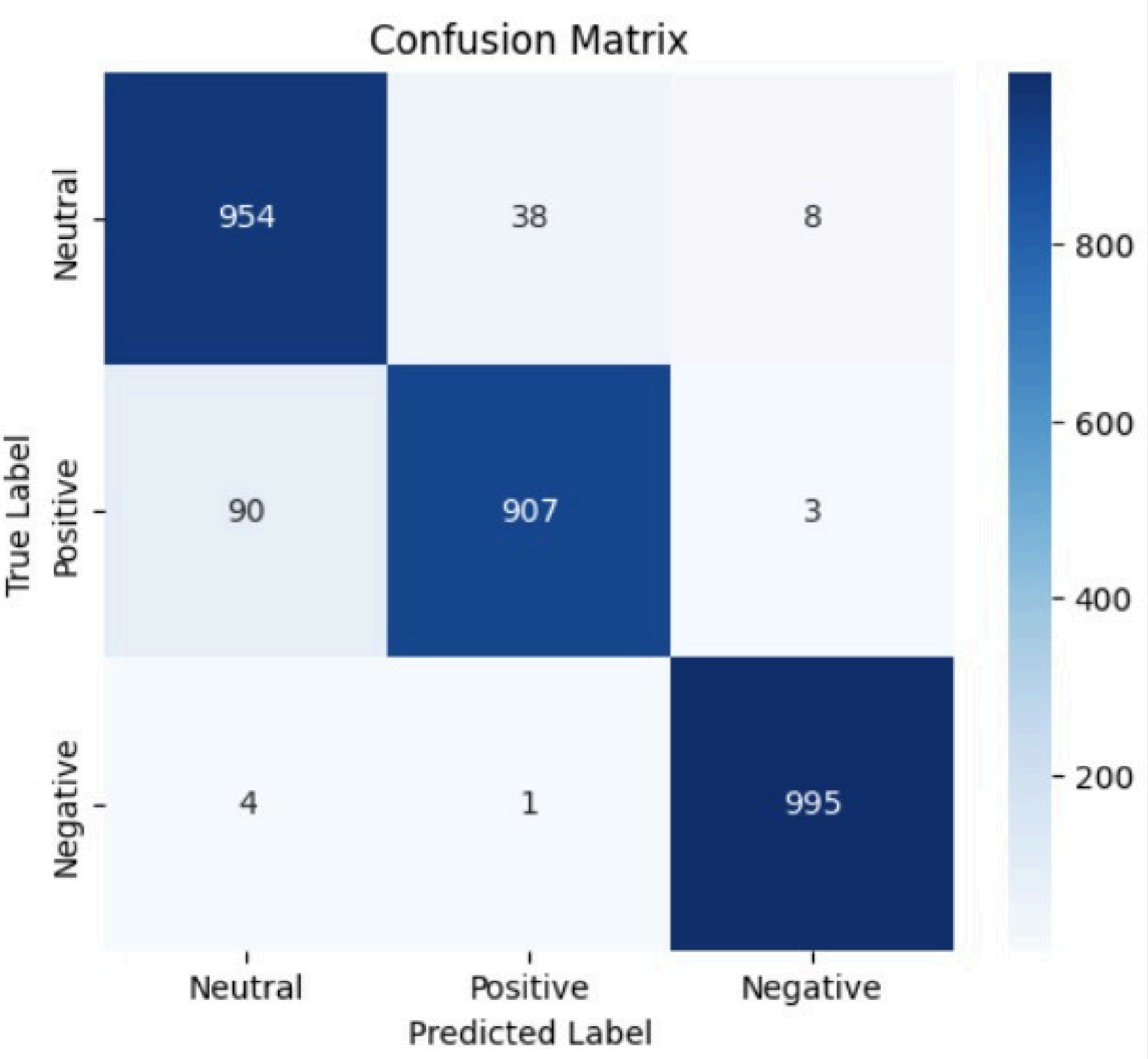
MODEL 2: BERT-BASE-UNCASED

	precision	recall	f1-score	support
Neutral	0.93	0.97	0.95	1000
Positive	0.97	0.92	0.95	1000
Negative	0.99	0.99	0.99	1000
accuracy			0.96	3000
macro avg	0.96	0.96	0.96	3000
weighted avg	0.96	0.96	0.96	3000

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.592000	0.308134	0.895333	0.895676	0.896385	0.895333
2	0.171300	0.215572	0.949000	0.949080	0.950297	0.949000
3	0.087400	0.213241	0.959000	0.958929	0.959660	0.959000
4	0.034900	0.215861	0.962333	0.962277	0.962924	0.962333

SENTIMENT CLASSIFICATION:

MODEL 3: ROBERTA-BASE



- **MODEL:** ROBERTA-BASE
- **DATASET:** 50,000 LABELED PRODUCT REVIEWS
- **CLASSES:** POSITIVE, NEGATIVE, NEUTRAL (MULTI-CLASS CLASSIFICATION)
- **ACCURACY:** 95% ON VALIDATION SET
- **PREPROCESSING:** TOKENIZATION WITH BERT TOKENIZER, LOWERCASING, SPECIAL TOKEN HANDLING
- **FINE-TUNING:** USED PRE-TRAINED BERT BASE MODEL WITH CLASSIFICATION HEAD
- **OPTIMIZER:** LEARNING RATE ADJUSTMENT
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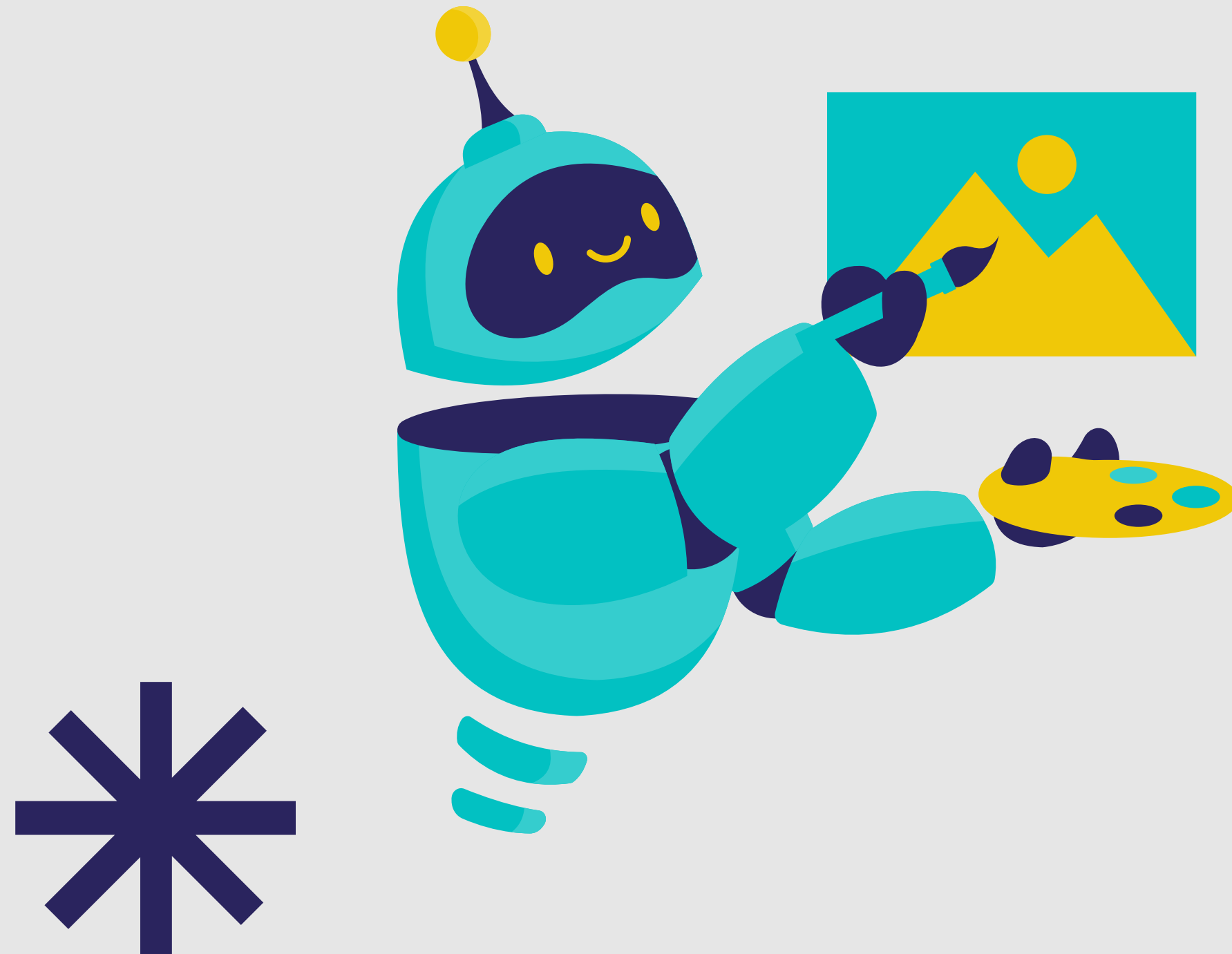
SENTIMENT CLASSIFICATION:

MODEL 3: ROBERTA-BASE

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.585900	0.325945	0.886333	0.887145	0.888814	0.886333
2	0.209900	0.256869	0.929333	0.929342	0.929632	0.929333
3	0.134200	0.282668	0.949333	0.949297	0.949309	0.949333
4	0.065300	0.290449	0.952000	0.951944	0.952715	0.952000

	precision	recall	f1-score	support
Neutral	0.91	0.95	0.93	1000
Positive	0.96	0.91	0.93	1000
Negative	0.99	0.99	0.99	1000
accuracy			0.95	3000
macro avg	0.95	0.95	0.95	3000
weighted avg	0.95	0.95	0.95	3000

BENIFETS



- CLUSTERING REVEALS UNDERLYING THEMES AND USER CONCERNS THAT AREN'T OBVIOUS AT FIRST GLANCE.
- SUMMARIZED FEEDBACK HELPS CUSTOMERS CHOOSE BETTER PRODUCTS AND BUSINESSES ADDRESS KEY ISSUES FASTER.
- ANALYZE THOUSANDS OF REVIEWS IN SECONDS—FAR BEYOND WHAT MANUAL ANALYSIS CAN HANDLE.

LOW MODEL ACCURACY

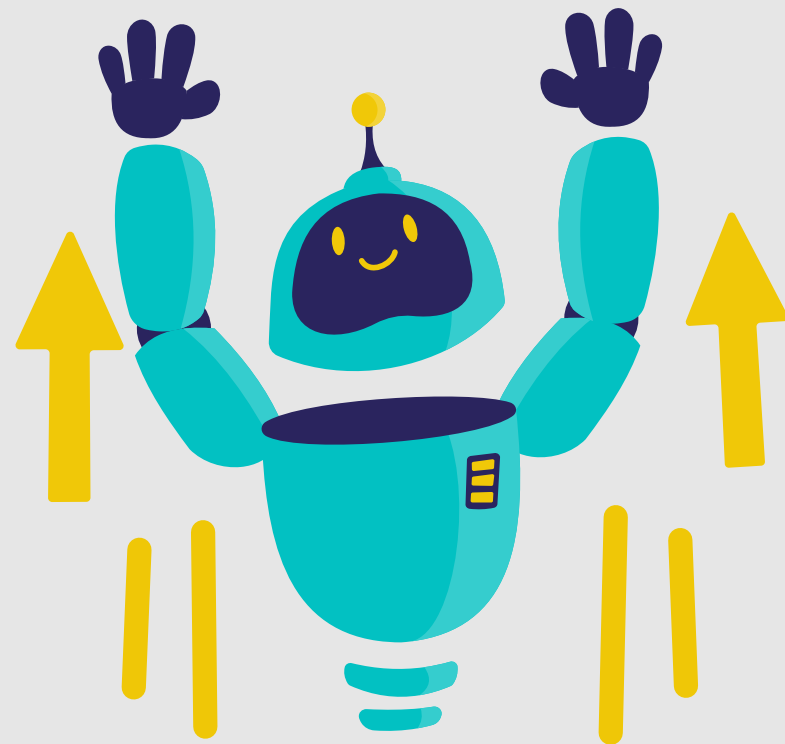
Challenge: Initial models achieved low accuracy due to noisy inputs and imbalanced classes

SOLUTION:

Cleaned and standardized review text

Balanced dataset with resampling techniques

Tuned hyperparameters (learning rate, batch size)



CHALLENGES

DEPLOYMENT AWS SAGEMAKER

Challenge: Packaging the model and setting up inference file

SOLUTION:

Try another deployment server but unfortunately a technical problem has appeared

DIFFICULTY INTERPRETING CLUSTERS

Challenge: K-Means clusters weren't clearly labeled or easy to understand

SOLUTION:

Analyzed top categories per cluster

Labeled clusters manually



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THANK YOU